Optical Flow-Based Dual Fisheye Stitching: Complete Development Guide

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Executive Summary

This document provides a comprehensive guide for implementing the **"optflow"** algorithm used by Insta360 SDK for seamless dual fisheye stitching. The implementation follows a two-phase approach: Python prototyping for algorithm validation, followed by C++ optimization for production performance.

The optical flow algorithm addresses the core challenge of creating seamless equirectangular panoramas from dual fisheye cameras by analyzing pixel-level motion in overlapping regions and using this information to guide intelligent seam detection and blending[1] [^22].

1. Algorithm Overview

1.1 What is the "OptFlow" Algorithm?

The **optical flow stitching algorithm** is Insta360's implementation of dense optical flow-based seam detection and blending for dual fisheye cameras[19][22]. Unlike traditional template-based stitching, optical flow analyzes actual pixel motion to create seamless transitions between the left and right fisheye images[^25].

Key Advantages:

- **Motion-aware stitching**: Handles moving objects crossing seam boundaries[22][25]

- **Parallax handling**: Better performance with near-field objects compared to template stitching[19] [28]
- **Adaptive seam placement**: Seam location adapts to scene content rather than fixed geometry[13][58]

Computational Hierarchy (Insta360's ranking)[^19]:

1. **AI Stitching** (highest quality, slowest)
2. **Optical Flow Stitching** (high quality, moderate speed) ← **Target Algorithm**
3. **Dynamic Stitching** (good quality, faster)
4. **Template Stitching** (basic quality, fastest)

1.2 Technical Foundation

The algorithm operates on the **optical flow equation**[87][96]:

$$f_x u + f_y v + f_t = 0$$

Where:

- (f_x, f_y) are spatial image gradients
- f_t is temporal gradient
- (u, v) is the motion vector to be determined

For dual fisheye stitching, we adapt this to handle overlapping regions between equirectangular projections of left and right fisheye images[13][61].

2. Implementation Pipeline

2.1 Complete Workflow

```
Input: Dual Fisheye Images (Left + Right)
    ↓
[Step 1] Fisheye Distortion Correction
    ↓
[Step 2] Equirectangular Projection
    ↓
[Step 3] Coarse Alignment
    ↓
[Step 4] Overlap Region Detection
    ↓
[Step 5] Optical Flow Computation
    ↓
[Step 6] Flow-Guided Seam Detection
    ↓
[Step 7] Multi-band Blending
    ↓
Output: Seamless Equirectangular Panorama
```

2.2 Algorithm Selection Strategy

For Python Prototype:

- **Primary**: Farneback Dense Optical Flow[87][161] (good balance of speed/quality)
- **Secondary**: Lucas-Kanade Pyramidal[87][92] (for feature tracking validation)

For C++ Production:

- **Primary**: Optimized Farneback with GPU acceleration
- **Fallback**: RLOF (Robust Local Optical Flow)[^87] for challenging lighting conditions

3. Python Implementation

3.1 Environment Setup

Dependencies:

```
pip install opencv-python&gt;=4.5.0
pip install numpy&gt;=1.21.0
pip install scipy&gt;=1.7.0
pip install matplotlib&gt;=3.3.0
```

System Requirements:

- Python ≥ 3.8
- RAM: 4-8GB minimum
- Processing: ~5-10 seconds per 4K frame

3.2 Core Implementation Classes

3.2.1 Fisheye Calibration and Unwarp

```
import cv2
import numpy as np
from typing import Tuple, Dict, Optional

class FisheyeProcessor:
    """Handles fisheye camera calibration and equirectangular conversion"""

    def __init__(self, camera_params: Dict):
        self.camera_matrix = np.array(camera_params['camera_matrix'])
        self.distortion_coeffs = np.array(camera_params['distortion_coeffs'])
        self.fov_degree = camera_params.get('fov_degree', 190)
        self.output_width = camera_params.get('output_width', 1920)
        self.output_height = camera_params.get('output_height', 960)

    def fisheye_to_equirectangular(self, fisheye_img: np.ndarray) -&gt; np.ndarray:
        """Convert fisheye image to equirectangular projection"""
```

```
        h, w = fisheye_img.shape[:2]

        # Create coordinate maps for equirectangular output
        eq_width, eq_height = self.output_width, self.output_height

        # Generate longitude and latitude maps
        lon_map = np.linspace(-np.pi, np.pi, eq_width)
        lat_map = np.linspace(np.pi/2, -np.pi/2, eq_height)

        # Create meshgrid
        lon_grid, lat_grid = np.meshgrid(lon_map, lat_map)

        # Convert to 3D coordinates
        x_3d = np.cos(lat_grid) * np.cos(lon_grid)
        y_3d = np.cos(lat_grid) * np.sin(lon_grid)
        z_3d = np.sin(lat_grid)

        # Project to fisheye coordinates
        r = np.sqrt(x_3d**2 + y_3d**2)
        theta = np.arctan2(r, z_3d)

        # Apply fisheye projection model
        r_fisheye = theta * (h / 2) / (self.fov_degree * np.pi / 360)
        phi = np.arctan2(y_3d, x_3d)

        # Convert to pixel coordinates
        fisheye_x = (w / 2) + r_fisheye * np.cos(phi)
        fisheye_y = (h / 2) + r_fisheye * np.sin(phi)

        # Remap the image
        map_x = fisheye_x.astype(np.float32)
        map_y = fisheye_y.astype(np.float32)

        equirect_img = cv2.remap(fisheye_img, map_x, map_y,
                                 cv2.INTER_LINEAR, cv2.BORDER_CONSTANT)

        return equirect_img

    def calibrate_from_checkerboard(self, checkerboard_images: list) -&gt; bool:
        """Auto-calibrate fisheye camera parameters"""
        # Implementation for automatic calibration using checkerboard patterns
        # This would use cv2.fisheye.calibrate() function
        pass
```

3.2.2 Optical Flow Stitcher

```
class OpticalFlowStitcher:
    """Main class implementing optical flow-based stitching"""

    def __init__(self, flow_algorithm: str = 'farneback'):
        self.flow_algorithm = flow_algorithm
        self.setup_flow_parameters()

    def setup_flow_parameters(self):
        """Initialize parameters for different optical flow algorithms"""
```

```
        if self.flow_algorithm == 'farneback':
            self.flow_params = {
                'pyr_scale': 0.5,      # Pyramid scale factor
                'levels': 3,           # Number of pyramid levels
                'winsize': 15,         # Averaging window size
                'iterations': 3,       # Number of iterations
                'poly_n': 5,          # Polynomial expansion neighborhood
                'poly_sigma': 1.2,    # Gaussian sigma for polynomial
                'flags': 0            # Operation flags
            }
        elif self.flow_algorithm == 'lucas_kanade':
            self.flow_params = {
                'winSize': (15, 15),
                'maxLevel': 2,
                'criteria': (cv2.TERM_CRITERIA_EPS | cv2.TERM_CRITERIA_COUNT, 10, 0.03)
            }

    def compute_optical_flow(self, img1: np.ndarray, img2: np.ndarray,
                             mask: Optional[np.ndarray] = None) -&gt; np.ndarray:
        """Compute optical flow between two images"""

        # Convert to grayscale for flow computation
        gray1 = cv2.cvtColor(img1, cv2.COLOR_BGR2GRAY) if len(img1.shape) == 3 else img1
        gray2 = cv2.cvtColor(img2, cv2.COLOR_BGR2GRAY) if len(img2.shape) == 3 else img2

        if self.flow_algorithm == 'farneback':
            flow = cv2.calcOpticalFlowFarneback(
                gray1, gray2, None,
                self.flow_params['pyr_scale'],
                self.flow_params['levels'],
                self.flow_params['winsize'],
                self.flow_params['iterations'],
                self.flow_params['poly_n'],
                self.flow_params['poly_sigma'],
                self.flow_params['flags']
            )

        elif self.flow_algorithm == 'lucas_kanade':
            # For Lucas-Kanade, we need feature points
            corners = cv2.goodFeaturesToTrack(gray1, 1000, 0.01, 10)
            if corners is not None:
                # Track features
                corners_new, status, error = cv2.calcOpticalFlowPyrLK(
                    gray1, gray2, corners, None, **self.flow_params
                )
                # Convert sparse flow to dense flow (interpolation needed)
                flow = self._sparse_to_dense_flow(corners, corners_new,
                                                  status, gray1.shape)
            else:
                flow = np.zeros((gray1.shape[^0], gray1.shape[^1], 2), dtype=np.float32)

        return flow

    def _sparse_to_dense_flow(self, corners_old, corners_new, status, img_shape):
        """Convert sparse Lucas-Kanade flow to dense flow field"""
        # Create dense flow field from sparse corner matches
```

```
        flow = np.zeros((img_shape[^0], img_shape[^1], 2), dtype=np.float32)

        # Filter good matches
        good_old = corners_old[status == 1]
        good_new = corners_new[status == 1]

        if len(good_old) &gt; 0:
            # Calculate displacement vectors
            displacement = good_new - good_old

            # Interpolate to create dense field (simple method)
            for i, (old_pt, disp) in enumerate(zip(good_old, displacement)):
                x, y = int(old_pt[^0]), int(old_pt[^1])
                if 0 &lt;= x &lt; img_shape[^1] and 0 &lt;= y &lt; img_shape[^0]:
                    flow[y, x] = disp

        # Apply Gaussian smoothing to create smoother dense field
        flow[:,:,0] = cv2.GaussianBlur(flow[:,:,0], (21, 21), 7)
        flow[:,:,1] = cv2.GaussianBlur(flow[:,:,1], (21, 21), 7)

        return flow

    def detect_overlap_region(self, left_eq: np.ndarray,
                              right_eq: np.ndarray) -&gt; Tuple[slice, slice]:
        """Detect overlapping region between left and right equirectangular images"""
        height, width = left_eq.shape[:2]

        # For dual fisheye, overlap is typically on left/right edges
        overlap_width = width // 6  # Assume ~16.7% overlap on each side

        # Left image: rightmost portion overlaps
        left_overlap_region = (slice(0, height), slice(width - overlap_width, width))

        # Right image: leftmost portion overlaps
        right_overlap_region = (slice(0, height), slice(0, overlap_width))

        return left_overlap_region, right_overlap_region

    def find_optimal_seam(self, flow: np.ndarray, overlap_mask: np.ndarray) -&gt; np.ndar
        """Find optimal seam line using optical flow information"""
        height, width = flow.shape[:2]

        # Calculate flow magnitude as a cost measure
        flow_magnitude = np.sqrt(flow[:,:,0]**2 + flow[:,:,1]**2)

        # Create cost matrix (higher flow = higher cost)
        cost_matrix = flow_magnitude.copy()

        # Apply overlap mask
        cost_matrix = cost_matrix * overlap_mask

        # Dynamic programming to find minimum cost path
        dp = np.zeros_like(cost_matrix)
        dp[0, :] = cost_matrix[0, :]

        # Fill DP table
```

```
        for i in range(1, height):
            for j in range(width):
                candidates = []
                # Check three possible previous positions
                for dj in [-1, 0, 1]:
                    if 0 &lt;= j + dj &lt; width:
                        candidates.append(dp[i-1, j + dj])
                dp[i, j] = cost_matrix[i, j] + min(candidates)

        # Backtrack to find optimal path
        seam_line = np.zeros(height, dtype=int)
        seam_line[-1] = np.argmin(dp[-1, :])

        for i in range(height - 2, -1, -1):
            prev_j = seam_line[i + 1]
            candidates = []
            for dj in [-1, 0, 1]:
                if 0 &lt;= prev_j + dj &lt; width:
                    candidates.append((dp[i, prev_j + dj], prev_j + dj))
            seam_line[i] = min(candidates)[^1]

        return seam_line

    def flow_guided_blending(self, left_img: np.ndarray, right_img: np.ndarray,
                             flow: np.ndarray, seam_line: np.ndarray,
                             blend_width: int = 50) -&gt; np.ndarray:
        """Perform flow-guided blending along the seam"""
        height, width = left_img.shape[:2]
        result = np.zeros_like(left_img)

        for i in range(height):
            seam_pos = seam_line[i]

            # Left side of seam: use left image
            if seam_pos &gt; blend_width:
                result[i, :seam_pos-blend_width] = left_img[i, :seam_pos-blend_width]

            # Right side of seam: use right image
            if seam_pos + blend_width &lt; width:
                result[i, seam_pos+blend_width:] = right_img[i, seam_pos+blend_width:]

            # Blending region: use flow-weighted combination
            blend_start = max(0, seam_pos - blend_width)
            blend_end = min(width, seam_pos + blend_width)

            for j in range(blend_start, blend_end):
                # Calculate blending weight based on distance to seam and flow magnitude
                dist_to_seam = abs(j - seam_pos)
                flow_mag = np.sqrt(flow[i, j, 0]**2 + flow[i, j, 1]**2)

                # Weight decreases with distance and increases with flow magnitude
                alpha = (blend_width - dist_to_seam) / blend_width
                alpha = alpha * (1 + 0.1 * flow_mag)  # Flow-based adjustment
                alpha = np.clip(alpha, 0, 1)

                if j &lt; seam_pos:
```

```
                    # Blend towards left image
                    result[i, j] = alpha * left_img[i, j] + (1 - alpha) * right_img[i, j]
                else:
                    # Blend towards right image
                    result[i, j] = (1 - alpha) * left_img[i, j] + alpha * right_img[i, j]

        return result

    def stitch_images(self, left_fisheye: np.ndarray, right_fisheye: np.ndarray,
                      fisheye_processor: FisheyeProcessor) -&gt; np.ndarray:
        """Complete stitching pipeline"""

        # Step 1: Convert fisheye to equirectangular
        left_eq = fisheye_processor.fisheye_to_equirectangular(left_fisheye)
        right_eq = fisheye_processor.fisheye_to_equirectangular(right_fisheye)

        # Step 2: Detect overlap regions
        left_overlap, right_overlap = self.detect_overlap_region(left_eq, right_eq)

        # Extract overlap regions
        left_overlap_img = left_eq[left_overlap]
        right_overlap_img = right_eq[right_overlap]

        # Step 3: Compute optical flow in overlap region
        flow = self.compute_optical_flow(left_overlap_img, right_overlap_img)

        # Step 4: Find optimal seam
        overlap_mask = np.ones(left_overlap_img.shape[:2], dtype=np.float32)
        seam_line = self.find_optimal_seam(flow, overlap_mask)

        # Step 5: Create full panorama canvas
        pano_height = left_eq.shape[^0]
        pano_width = left_eq.shape[^1] + right_eq.shape[^1] - left_overlap_img.shape[^1]
        panorama = np.zeros((pano_height, pano_width, 3), dtype=np.uint8)

        # Place left image
        left_end = left_eq.shape[^1] - left_overlap_img.shape[^1]
        panorama[:, :left_end] = left_eq[:, :left_end]

        # Place right image
        right_start = left_end + left_overlap_img.shape[^1]
        panorama[:, right_start:] = right_eq[:, left_overlap_img.shape[^1]:]

        # Step 6: Blend overlap region
        blended_overlap = self.flow_guided_blending(
            left_overlap_img, right_overlap_img, flow, seam_line
        )
        panorama[:, left_end:left_end + left_overlap_img.shape[^1]] = blended_overlap

        return panorama
```

3.3 Usage Example

```
def main():
    # Camera calibration parameters (example for Insta360 X4)
    camera_params = {
        'camera_matrix': [[1000, 0, 960], [0, 1000, 960], [0, 0, 1]],
        'distortion_coeffs': [-0.1, 0.05, 0, 0],
        'fov_degree': 195,
        'output_width': 1920,
        'output_height': 960
    }

    # Initialize processors
    fisheye_proc = FisheyeProcessor(camera_params)
    stitcher = OpticalFlowStitcher('farneback')

    # Load dual fisheye images
    left_fisheye = cv2.imread('left_fisheye.jpg')
    right_fisheye = cv2.imread('right_fisheye.jpg')

    # Perform stitching
    panorama = stitcher.stitch_images(left_fisheye, right_fisheye, fisheye_proc)

    # Save result
    cv2.imwrite('panorama_optflow.jpg', panorama)

    print("Optical flow stitching completed!")

if __name__ == "__main__":
    main()
```

4. C++ Production Implementation

4.1 Performance Requirements

Target Specifications:

- **Resolution**: 4K+ (3840×1920 and above)
- **Processing Time**: 100-500ms per frame
- **Memory Usage**: 1-2GB RAM maximum
- **Throughput**: 10-30 FPS for real-time video processing

4.2 Optimization Strategies

4.2.1 CUDA Acceleration

```
#include &lt;opencv2/opencv.hpp&gt;
#include &lt;opencv2/cudaimgproc.hpp&gt;
#include &lt;opencv2/cudaarithm.hpp&gt;
#include &lt;opencv2/cudaoptflow.hpp&gt;

class CudaOpticalFlowStitcher {
private:
    cv::cuda::GpuMat d_frame1, d_frame2, d_flow;
    cv::Ptr&lt;cv::cuda::FarnebackOpticalFlow&gt; farneback;

public:
    CudaOpticalFlowStitcher() {
        // Initialize CUDA Farneback optical flow
        farneback = cv::cuda::FarnebackOpticalFlow::create(
            3,      // numLevels
            0.5,    // pyrScale
            false,  // fastPyramids
            15,     // winSize
            3,      // numIters
            5,      // polyN
            1.2,    // polySigma
            0       // flags
        );
    }

    cv::Mat computeOpticalFlowGPU(const cv::Mat&amp; frame1, const cv::Mat&amp; frame2) {
        // Upload to GPU
        d_frame1.upload(frame1);
        d_frame2.upload(frame2);

        // Compute optical flow on GPU
        farneback-&gt;calc(d_frame1, d_frame2, d_flow);

        // Download result
        cv::Mat flow;
        d_flow.download(flow);

        return flow;
    }
};
```

4.2.2 Multi-threading Architecture

```
#include &lt;thread&gt;
#include &lt;future&gt;
#include &lt;queue&gt;
#include &lt;mutex&gt;

class ThreadedOpticalFlowStitcher {
private:
    std::queue&lt;cv::Mat&gt; input_queue;
    std::queue&lt;cv::Mat&gt; output_queue;
```

```
    std::mutex queue_mutex;
    bool processing_active;

    // Thread pool for parallel processing
    std::vector&lt;std::thread&gt; worker_threads;
    const int num_threads;

public:
    ThreadedOpticalFlowStitcher(int threads = std::thread::hardware_concurrency())
        : num_threads(threads), processing_active(true) {

        // Launch worker threads
        for (int i = 0; i &lt; num_threads; ++i) {
            worker_threads.emplace_back(&amp;ThreadedOpticalFlowStitcher::workerFunction,
        }
    }

    void workerFunction() {
        while (processing_active) {
            cv::Mat input_frame;

            // Get work from queue
            {
                std::lock_guard&lt;std::mutex&gt; lock(queue_mutex);
                if (!input_queue.empty()) {
                    input_frame = input_queue.front();
                    input_queue.pop();
                }
            }

            if (!input_frame.empty()) {
                // Process frame
                cv::Mat result = processFrame(input_frame);

                // Add to output queue
                {
                    std::lock_guard&lt;std::mutex&gt; lock(queue_mutex);
                    output_queue.push(result);
                }
            }

            std::this_thread::sleep_for(std::chrono::milliseconds(1));
        }
    }

    cv::Mat processFrame(const cv::Mat&amp; dual_fisheye) {
        // Implementation of complete processing pipeline
        // This would include all steps from Python version but optimized
        return cv::Mat();
    }
};
```

4.3 Memory Optimization

```
class MemoryOptimizedStitcher {
private:
    // Pre-allocated buffers to avoid dynamic allocation
    cv::Mat buffer_left_eq, buffer_right_eq;
    cv::Mat buffer_overlap_left, buffer_overlap_right;
    cv::Mat buffer_flow, buffer_result;

    // Lookup tables for fisheye unwrapping (computed once)
    cv::Mat map_x_left, map_y_left;
    cv::Mat map_x_right, map_y_right;

public:
    void precomputeLookupTables(const CameraParams&amp; params) {
        // Pre-compute remap tables for fisheye to equirectangular conversion
        // This avoids recalculating trigonometric functions for each frame

        int eq_width = params.output_width;
        int eq_height = params.output_height;

        map_x_left.create(eq_height, eq_width, CV_32FC1);
        map_y_left.create(eq_height, eq_width, CV_32FC1);

        // Fill lookup tables with pre-computed coordinates
        for (int y = 0; y &lt; eq_height; ++y) {
            for (int x = 0; x &lt; eq_width; ++x) {
                // Convert equirectangular coordinates to fisheye coordinates
                double longitude = (x / double(eq_width)) * 2.0 * M_PI - M_PI;
                double latitude = (y / double(eq_height)) * M_PI - M_PI/2.0;

                // 3D sphere coordinates
                double x3d = cos(latitude) * cos(longitude);
                double y3d = cos(latitude) * sin(longitude);
                double z3d = sin(latitude);

                // Project to fisheye
                double r = sqrt(x3d*x3d + y3d*y3d);
                double theta = atan2(r, z3d);
                double phi = atan2(y3d, x3d);

                double r_fisheye = theta * (params.fisheye_height / 2.0) / (params.fov_ra

                map_x_left.at&lt;float&gt;(y, x) = params.fisheye_center_x + r_fisheye *
                map_y_left.at&lt;float&gt;(y, x) = params.fisheye_center_y + r_fisheye *
            }
        }
    }

    cv::Mat stitchOptimized(const cv::Mat&amp; left_fisheye, const cv::Mat&amp; right_fis
        // Use pre-computed lookup tables for fast remapping
        cv::remap(left_fisheye, buffer_left_eq, map_x_left, map_y_left,
                  cv::INTER_LINEAR, cv::BORDER_CONSTANT);
        cv::remap(right_fisheye, buffer_right_eq, map_x_right, map_y_right,
                  cv::INTER_LINEAR, cv::BORDER_CONSTANT);
```

```
        // Continue with optical flow processing using pre-allocated buffers
        // ... rest of implementation

        return buffer_result;
    }
};
```

5. Parameter Tuning and Optimization

5.1 Farneback Algorithm Parameters

Parameter	Default	Range	Tuning Guidelines
`pyr_scale`	0.5	0.1-0.9	**Lower values** (0.3-0.4) for fine details, **higher values** (0.6-0.8) for speed
`levels`	3	1-5	**More levels** (4-5) for large motions, **fewer levels** (2-3) for speed
`winsize`	15	5-25	**Larger windows** (20-25) for noisy images, **smaller windows** (10-15) for sharp details
`iterations`	3	1-10	**More iterations** (5-8) for accuracy, **fewer iterations** (2-3) for speed
`poly_n`	5	3-7	**Higher values** (7) for smooth regions, **lower values** (3-5) for textured areas
`poly_sigma`	1.2	0.5-2.0	**Lower values** (0.8-1.0) for sharp features, **higher values** (1.5-2.0) for noise

5.2 Scene-Specific Optimization

For Indoor Scenes (controlled lighting):

```
indoor_params = {
    'pyr_scale': 0.6,     # Higher for efficiency
    'levels': 3,          # Standard
    'winsize': 12,        # Smaller for sharp details
    'iterations': 2,      # Fewer for speed
    'poly_n': 5,         # Standard
    'poly_sigma': 1.0    # Lower for sharp features
}
```

For Outdoor Scenes (variable lighting, moving objects):

```
outdoor_params = {
    'pyr_scale': 0.4,     # Lower for fine motion detection
    'levels': 4,          # More levels for large motions
    'winsize': 18,        # Larger for robustness
    'iterations': 4,      # More for accuracy
    'poly_n': 5,         # Standard
    'poly_sigma': 1.4    # Higher for noise handling
}
```

For High-Speed Motion (sports, action):

```
motion_params = {
    'pyr_scale': 0.3,     # Lower for large displacements
    'levels': 5,          # Maximum levels
    'winsize': 20,        # Large window for tracking
    'iterations': 3,      # Balanced
    'poly_n': 7,         # Higher for smoothing
    'poly_sigma': 1.6    # Higher for motion blur
}
```

6. Quality Assessment and Validation

6.1 Stitching Quality Metrics

```
def evaluate_stitching_quality(panorama: np.ndarray, seam_line: np.ndarray) -&gt; Dict:
    """Evaluate stitching quality with multiple metrics"""

    metrics = {}

    # 1. Seam Visibility Score
    seam_gradient = calculate_seam_gradient(panorama, seam_line)
    metrics['seam_visibility'] = np.mean(seam_gradient)

    # 2. Color Consistency
    color_diff = calculate_color_difference_across_seam(panorama, seam_line)
    metrics['color_consistency'] = 1.0 / (1.0 + color_diff)

    # 3. Structural Similarity (SSIM) in overlap regions
    ssim_score = calculate_overlap_ssim(panorama, seam_line)
    metrics['structural_similarity'] = ssim_score

    # 4. Overall Quality Score (weighted combination)
    metrics['overall_quality'] = (
        0.4 * (1.0 - metrics['seam_visibility']) +
        0.3 * metrics['color_consistency'] +
        0.3 * metrics['structural_similarity']
    )

    return metrics
```

6.2 Performance Benchmarking

```
def benchmark_performance(stitcher, test_images: list) -&gt; Dict:
    """Benchmark stitching performance across different image types"""

    results = {
        'processing_times': [],
        'memory_usage': [],
        'quality_scores': []
    }
```

```
    for left_img, right_img in test_images:
        start_time = time.time()

        # Monitor memory usage
        memory_before = psutil.Process().memory_info().rss

        # Perform stitching
        panorama = stitcher.stitch_images(left_img, right_img)

        processing_time = time.time() - start_time
        memory_after = psutil.Process().memory_info().rss

        results['processing_times'].append(processing_time)
        results['memory_usage'].append(memory_after - memory_before)

        # Evaluate quality (if ground truth available)
        quality_score = evaluate_stitching_quality(panorama)
        results['quality_scores'].append(quality_score['overall_quality'])

    # Calculate statistics
    results['avg_processing_time'] = np.mean(results['processing_times'])
    results['avg_memory_usage'] = np.mean(results['memory_usage']) / (1024*1024)  # MB
    results['avg_quality'] = np.mean(results['quality_scores'])

    return results
```

7. Integration with Insta360 Workflow

7.1 .insv File Processing

```
def process_insv_file(insv_path: str, output_dir: str):
    """Process Insta360 .insv file using optical flow stitching"""

    # Extract dual fisheye frames using Insta360 SDK or FFmpeg
    cmd = f'ffmpeg -i "{insv_path}" -vf "crop=1920:1920:0:0" -q:v 2 "{output_dir}/left_%0
    os.system(cmd)

    cmd = f'ffmpeg -i "{insv_path}" -vf "crop=1920:1920:1920:0" -q:v 2 "{output_dir}/righ
    os.system(cmd)

    # Initialize stitcher
    camera_params = get_insta360_x4_params()  # Load calibration
    fisheye_proc = FisheyeProcessor(camera_params)
    stitcher = OpticalFlowStitcher('farneback')

    # Process each frame pair
    left_files = sorted(glob.glob(f"{output_dir}/left_*.jpg"))
    right_files = sorted(glob.glob(f"{output_dir}/right_*.jpg"))

    for left_file, right_file in zip(left_files, right_files):
        left_img = cv2.imread(left_file)
        right_img = cv2.imread(right_file)
```

```
        panorama = stitcher.stitch_images(left_img, right_img, fisheye_proc)

        # Save stitched result
        frame_num = os.path.basename(left_file).split('_')[^1].split('.')[^0]
        output_path = f"{output_dir}/panorama_{frame_num}.jpg"
        cv2.imwrite(output_path, panorama)
```

7.2 Comparison with Insta360 SDK Results

```
def compare_with_insta360_sdk(test_cases: list):
    """Compare our optical flow implementation with Insta360 SDK results"""

    comparison_results = []

    for test_case in test_cases:
        # Our implementation
        our_result = our_stitcher.stitch_images(
            test_case['left_fisheye'],
            test_case['right_fisheye']
        )

        # Insta360 SDK result (if available)
        sdk_result = test_case.get('insta360_sdk_result')

        if sdk_result is not None:
            # Calculate similarity metrics
            ssim_score = calculate_ssim(our_result, sdk_result)
            psnr_score = calculate_psnr(our_result, sdk_result)

            comparison_results.append({
                'test_case': test_case['name'],
                'ssim': ssim_score,
                'psnr': psnr_score,
                'visual_similarity': ssim_score &gt; 0.85  # Threshold for acceptable qua
            })

    return comparison_results
```

8. Troubleshooting and Common Issues

8.1 Common Problems and Solutions

Problem	Symptoms	Solution
Visible Seam Lines	Obvious stitching artifacts	Adjust `poly_sigma` (increase), optimize seam detection algorithm
Motion Blur in Flow	Incorrect flow vectors	Reduce `pyr_scale`, increase `levels`
Poor Performance	Slow processing	Use GPU acceleration, optimize memory allocation
Misalignment	Objects don't line up	Check camera calibration, improve coarse alignment

Problem	Symptoms	Solution
Color Discontinuity	Color shifts across seam	Implement color correction, adjust blending weights

8.2 Debug Visualization Tools

```
def visualize_optical_flow(flow: np.ndarray) -&gt; np.ndarray:
    """Create HSV visualization of optical flow"""
    magnitude, angle = cv2.cartToPolar(flow[..., 0], flow[..., 1])

    hsv = np.zeros((flow.shape[^0], flow.shape[^1], 3), dtype=np.uint8)
    hsv[..., 0] = angle * 180 / np.pi / 2
    hsv[..., 1] = 255
    hsv[..., 2] = cv2.normalize(magnitude, None, 0, 255, cv2.NORM_MINMAX)

    flow_vis = cv2.cvtColor(hsv, cv2.COLOR_HSV2BGR)
    return flow_vis

def debug_seam_detection(left_overlap: np.ndarray, right_overlap: np.ndarray,
                         flow: np.ndarray, seam_line: np.ndarray) -&gt; np.ndarray:
    """Create debug visualization showing seam detection process"""
    debug_img = np.hstack([left_overlap, right_overlap])

    # Draw seam line
    for i, seam_pos in enumerate(seam_line):
        cv2.circle(debug_img, (seam_pos + left_overlap.shape[^1], i), 2, (0, 255, 0), -1)

    # Overlay flow visualization
    flow_vis = visualize_optical_flow(flow)
    debug_img = cv2.addWeighted(debug_img, 0.7,
                                cv2.hstack([flow_vis, flow_vis]), 0.3, 0)

    return debug_img
```

9. Future Enhancements

9.1 Deep Learning Integration

Consider integrating modern deep learning approaches for enhanced quality:

- **PWC-Net** or **RAFT** for more accurate optical flow[146][147]
- **Neural seam detection** using learned features rather than hand-crafted metrics
- **Temporal consistency** networks for video stitching

9.2 Real-time Optimization

For real-time applications:

- **ROI-based processing**: Only compute optical flow in critical regions
- **Adaptive quality**: Reduce quality parameters based on processing load

- **Frame skipping**: Process keyframes and interpolate intermediate results

9.3 Advanced Calibration

- **Online calibration**: Adapt camera parameters based on scene content
- **Multi-camera support**: Extend to more than dual fisheye setups
- **Distortion learning**: Use neural networks to learn camera-specific distortions

10. Conclusion

The optical flow-based stitching algorithm ("optflow") provides superior quality compared to template-based methods by intelligently analyzing pixel motion in overlap regions [1] [13][22]. This implementation guide provides both prototype Python code for algorithm development and optimized C++ strategies for production deployment.

Key Success Factors:

1. **Proper calibration**: Accurate fisheye camera parameters are critical[124][126]
2. **Parameter tuning**: Scene-specific optimization of optical flow parameters[87][161]
3. **Performance optimization**: GPU acceleration and memory management for real-time processing[^88]
4. **Quality validation**: Continuous testing against reference implementations[19][25]

By following this comprehensive guide, developers can successfully replicate and potentially improve upon Insta360's "optflow" algorithm for seamless dual fisheye stitching applications.

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